

Prodromos Gerakios, PhD

Location: Zurich, Switzerland

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Summary

Senior Software Engineer with 15+ years' experience and a PhD in Computer Science (programming languages, type systems, concurrency). Currently at **Google DeepMind (Zurich)** working on cross-platform agentic workflows for the Gemini app. Deep expertise in large-scale distributed systems, cloud/AI platforms, real-time ML inference, and production reliability. Track record of shipping high-impact infrastructure across Google, Workday, and European research projects, with published research in top-tier programming languages venues (OOPSLA, GPCE, SCP).

Key Achievements

- > Cut LLM operational costs by **4x** on Google DeepMind's cross-platform agentic UI-actuation platforms by consolidating Web, iOS, macOS, and Android stacks.
- > Reduced Gemini-app service error rates from ~2% to **0.016% (125x improvement)** via a 3-stage availability program (quota modeling, smart model routing, dynamic failover with jitter back-offs).
- > Architected a real-time fraud prediction service at Google with **sub-20 ms latency at 1,000 QPS**.
- > Delivered a **30% reduction in GCU/RAM** on Google Search batch pipelines (equivalent to ~1.5 nominal employee salaries saved annually).
- > Accelerated test development by **8x** at Google CCAIP by leading a small team on a new DSL for deterministic test-code generation from semi-structured specs.
- > Improved distributed job execution time by **80%** at Workday's Cloud Master scheduler.
- > Published **10 peer-reviewed papers** in top-tier programming-languages venues (OOPSLA, GPCE, SCP, TLDI, PLOS).

Work Experience

♦ Senior Software Engineer — Google DeepMind, Zurich

Jan 2026 – Present · Infrastructure & AI Engineer, Cross-Platform Agentic Workflows for the Gemini App

- > Led architectural consolidation of AI-driven UI-actuation platforms across Web, iOS, macOS, and Android, **cutting LLM operational costs by 4x** and eliminating redundant engineering builds.
- > Designed and provisioned foundational cloud infrastructure, billing models, and security perimeters for high-scale AI testing workloads, unblocking critical mobile and desktop testing workstreams.
- > Engineered a 3-stage availability program (quota modeling, smart model routing, dynamic failover with jitter back-offs), **reducing service error rates from ~2% to 0.016% (125x improvement)**.
- > Developed core Generative AI testing primitives: Gemini-driven browser controls, pluggable planner interfaces, and a novel **vision-LLM value-extraction pipeline** for autonomous visual state validation.
- > Architected a highly observable cloud proxy with baked-in structured logging and tracing, cutting incident root-cause analysis from hours to minutes.
- > Shipped an atomic blue/green deployment tool with OIDC verification for safe high-velocity iteration.
- > Led complex security and interoperability integrations, including a secure proxy bridge for enterprise mobile workflows and an auditable Multi-Party Authorization (MPA) bypass to eliminate system latency.

Technologies: Generative AI (Gemini), Vision-LLMs, GCP, Cloud Run, Terraform, Distributed Systems, CI/CD, Secret Manager, OIDC, MPA, WebDriver.

♦ Senior Software Engineer — Google, Zurich · Contact Center AI Platform (Applied AI)

Nov 2022 – Dec 2025

- > Led a small team building deterministic test-code generation from semi-structured specs; developed a new DSL that **accelerated test development time by 8x**.
- > Led development of a shared visual-AI quality framework accepting natural-language instructions to crawl any visual interface.
- > Led Vo design and implementation of the Pindrop real-time voice authentication and fraud detection integration for Google's Contact Center AI Platform; aligned Pindrop, telephony carriers (Telnyx, Twilio), Ujet, and Google teams to enable real-time voice authentication with virtual agents; managed bidirectional voice streams and metadata (DTMF, ANI, DNIS) under stringent latency and accuracy requirements.
- > Designed and enforced cloud security posture for CCAIP via VPC Service Controls, Private Service Connect, and Organization Policies preventing data exfiltration; became point-of-contact for social-channels teams on VPC-SC compliance.

Technologies: GCP (GKE, Kubernetes, Load Balancing), Terraform, Golang, AI chatbots, GTP, RTP, SIP, OnePlatform, CCFE, PSC (L4/L7, endpoints, backends), IAM, Chemist, Inception.

♦ Senior Software Engineer — Google, Zurich · Financial Services (Applied AI)

Jan 2021 – Nov 2022 · Infrastructure Engineer, Real-time Fraud Prediction

- Developed and shipped a real-time fraud prediction service achieving **sub-20 ms latency at 1,000 QPS**.
- Pioneered an "Active MemCache" architecture with custom aggregation trees, co-locating data and computation to resolve critical performance bottlenecks in feature retrieval.
- Enhanced production reliability by resolving Seccomp/GKE issues, developing Dataflow performance dashboards, and optimizing capacity planning.
- Led the AxT PoC, productionized the Active Feature Cache service, and integrated Cloud Spanner into AxT.

Technologies: Borg, Golang, Apps Framework, CCFE, OnePlatform, BigQuery, GKE, MLOps, Kubeflow, gRPC, AxT, VPC-SC.

♦ Software Engineer — Google, Zurich · RefX (Search)

Jul 2018 – Dec 2020 · Data Science & Infrastructure Engineer, Entity Names

- Key contributor to knowledge-panel display for web results via semantic query interpretation, impacting Google Search, Assistant, and YouTube.
- Optimized batch pipelines: **30% reduction in GCU/RAM** (equivalent to ~1.5 nominal employee salaries saved annually).
- Improved sensitive-entity alias precision, refined annotator-level smearing, achieved substantial storage savings on alpha name tables, and re-enabled Chain Entity Name Propagation for more accurate tracking.

Technologies: C++, Big Data, Borg, Protobuf, Information Retrieval, Flume, SSTables, Knowledge Graph.

♦ Software Development Engineer III — Workday, Dublin

May 2016 – Jun 2018 · Cloud Master (Job Scheduler)

- Delivered Workday's next-generation distributed job scheduler for its private cloud.
- **80% improvement in job execution time** via distributed scheduling algorithm enhancements.
- Implemented Zero-Downtime Deployment (ZDT) for the service.

Technologies: Scala, Akka (clustering, remoting, actors), Zookeeper, Kafka, Java 8, MySQL, Spring, CDI, Hibernate, Docker, Jenkins.

♦ **Software Engineer — Advantage, Athens · Agiliti E-Banking (Banking SaaS)**

Jun 2015 – Apr 2016

- Full-stack Java web developer for a UK retail e-banking SaaS (in partnership with Fiserv) covering accounts, loans, payments, and e-statement reporting.

Technologies: JEE7, AngularJS, PostgreSQL, WildFly, Infinispan, Hibernate.

♦ **Software Engineer — Cognition, Athens**

Mar 2014 – Feb 2015 · Cosmote Romania Portal Redesign · ATG Germanos E-commerce

- Delivered high-traffic e-commerce platforms for major telecom providers and retailers on Oracle ATG Web Commerce.

Technologies: JEE6, Oracle ATG, WebLogic, Oracle 11.

♦ **Senior Researcher — University of Athens · MorphPL Project**

Dec 2012 – Mar 2014

- Developed Java compilers with advanced metaprogramming (compile-time reflection, prototype-based inheritance).
- Published in **OOPSLA** and **GPCE** (see Publications).
- Co-advised PhD students at PLAST lab.

Technologies: Java, Compilers, Metaprogramming.

♦ **Software Engineer — GRNET (Greek Research Network), Athens**

May 2012 – Dec 2012 · Aquarium billing backend for Okeanos Cloud IaaS

Technologies: Python, Scala, Akka, MongoDB, RabbitMQ.

♦ **Software Engineer — Hellenic Army Computer Support Center (obligatory service)**

Sept 2011 – May 2012

- Developed HR software.

Technologies: Java, Swing, JBoss, Oracle GI.

♦ **Software Engineer — GRNET, Athens · European research projects**

Nov 2006 – May 2009 · GridCC (grid computing) · Geant/PerfSONAR (network monitoring)

- Designed and delivered C++/Java web services for real-time control of distributed instrumentation.

Technical Skills

♦ Expert (5+ years of production experience)

Category	Technologies
Languages	Java, Golang, C++, Python, Scala
Cloud & Platforms	Google Cloud Platform (GCP), Kubernetes, Borg, Docker
Architecture	Distributed Systems, Microservices, Event-Driven Architecture, API Design
Databases	BigQuery, PostgreSQL, MySQL

♦ Proficient (2-5 years of experience)

Category	Technologies
Infrastructure	Terraform, Jenkins, GKE, Cloud Run
Messaging	Kafka, RabbitMQ, gRPC
AI / ML	Generative AI (Gemini), Vision-LLMs, MLOps, Kubeflow
Databases	Cloud Spanner, MongoDB
Security	VPC-SC, PSC, IAM, OIDC, Secret Manager

♦ Familiar (1-2 years or non-production experience)

Category	Technologies
Languages	Kotlin, JavaScript
Frameworks	Spring, Hibernate, Akka, AngularJS

Education

♦ PhD in Computer Science — National Technical University of Athens

Jan 2007 – Sept 2012 - Thesis: Static Safety Guarantees for Concurrent Programming Languages - Published in OOPSLA, GPCE, SCP, TLDI, PLOS, EPTCS, FTfJP.

♦ **International MBA, E-Business Specialization — Athens University of Business and Economics**

Aug 2005 – Nov 2006

♦ **BA (Hons), Software Engineering — University of Edinburgh**

Sept 2001 – Jun 2005 · First class with distinction - **University of Edinburgh Class Medal** for academic excellence (ranked 1st in Software Engineering class of 2005). - **British Computer Society Prize** for best final-year project (2005). - **Microsoft Prize** for a decentralized P2P file system (2004).

Selected Publications

1. **Gerakios, P.**, Fourtounis, G., Smaragdakis, Y. Foo: A Minimal Modern OO Calculus. 17th Workshop on Formal Techniques for Java-like Programs (FTfJP), Prague, 2015.
2. **Gerakios, P.**, Papaspyrou, N., Sagonas, K. Static Safety Guarantees for a Low-level Multithreaded Language with Regions. Science of Computer Programming, vol. 80(B), pp. 223–263, 2014.
3. **Gerakios, P.**, Biboudis, A., Smaragdakis, Y. Forsaking Inheritance: Supercharged Delegation in DelphJ. OOPSLA '13, pp. 233–252, ACM, 2013.
4. **Gerakios, P.**, Biboudis, A., Smaragdakis, Y. Reified Type Parameters using Java Annotations. GPCE, pp. 61–64, 2013.
5. **Gerakios, P.**, Papaspyrou, N., Sagonas, K. A Type and Effect System for Deadlock Avoidance in Low-level Languages. ACM SIGPLAN Workshop on Types in Languages Design and Implementation (TLDI), pp. 15–28, 2011.
6. **Gerakios, P.**, Papaspyrou, N., Sagonas, K., Vekris, P. Dynamic Deadlock Avoidance in Systems Code Using Statically Inferred Effects. 6th Workshop on Programming Languages and Operating Systems (PLOS '11), ACM, 2011.
7. Sagonas, K., **Gerakios, P.**, Papaspyrou, N. A Type System for Unstructured Locking that Guarantees Deadlock Freedom without Imposing a Lock Ordering. EPTCS vol. 69, pp. 44–58, 2011.
8. **Gerakios, P.**, Papaspyrou, N., Sagonas, K. A Concurrent Language with a Uniform Treatment of Regions and Locks. EPTCS vol. 17, pp. 79–93, 2010.
9. **Gerakios, P.**, Papaspyrou, N., Sagonas, K. Race-free and Memory-safe Multithreading: Design and Implementation in Cyclone. ACM SIGPLAN Workshop on Types in Languages Design and Implementation (TLDI), pp. 15–26, 2010.
10. Papakyriakou, M.A., **Gerakios, P.E.**, Papaspyrou, N.S. A Mechanized Proof of Type Safety for the Polymorphic Lambda Calculus with References. Panhellenic Logic Symposium, 2007.

Languages

Language	Proficiency
Greek	Native
English	Fluent